Shayan Salehi

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Education

M.Sc. *Artificial Intelligence* Sep 2025 – Present
Sharif University of Technology, Tehran
GPA **18.56**/20

B.Sc. *Computer Engineering* Sep 2020 – Feb 2025
Sharif University of Technology, Tehran
GPA **18.90**/20 | Major GPA **18.96**/20

Research Interests

Vision Language Models | Tool-Augmented Language Models | Biomedical Informatics
Human-Computer Interaction | AI in Healthcare

Publications

[1] **Learning from Disagreement: Maximum Divergence Knowledge Distillation**
A. Jafari, P. Fashi, M. Rezagholizadeh, H. Golmankhaneh, **S. Salehi**, V. Appia, E. Barsoum, A. Ghodsi
NeurIPS 2026 Conference Published

[2] **The Compute Floor of Data Mixing: When Loss-Based Reweighters Start to Drain Easy Synthetic Text**
S. Salehi, I. Mohammadi
COLM 2026 Workshop on Methods and Opportunities at Small Scale (MOSS) Published

[3] **PrivacyPeek: Showing Users What Their AI Chats Send Out, and When It Crosses a Line**
I. Mohammadi, **S. Salehi**
USIT 2026 Poster Session Published

[4] **Structure Matters: Evaluating Multi-Agents Orchestration in Generative Therapeutic Chatbots**
S. Elahimanesh, M. Mohammadkhani, S. Zahedi Movahed, M. Abootorabi, **S. Salehi**, A. Edalat
ACM CUI 2026 Conference Published

[5] **Folk Theories of Federated Moderation: How Users Make Sense of Cross-Instance Action**
I. Mohammadi, **S. Salehi**
CSCW 2026 Poster Session Published

[6] **Understanding Students' Rationales for Cheating with AI: An Instructor Perspective**
I. Mohammadi, S. Rismanchian, **S. Salehi**
ICLS 2026 Conference Poster Session Published

[7] **From Words and Exercises to Wellness: Farsi Chatbot for Self-Attachment**
S. Elahimanesh, **S. Salehi**, S. Zahedi Movahed, L. Alazraki, R. Hu, A. Edalat
EAI Endorsed Transactions on Pervasive Health and Technology (journal) Under review

Honors & Awards

1st **NLP Tech Olympics** Oct 2025
Among 2,100+ teams (top 0.05%)

10th **National Student Olympiad** Aug 2024
Among 1,200+ candidates (top 0.8%)

120 **NODET Entrance Exam** Sep 2014
Top 120 of 11,000+ (top 1.1%)

16th **National M.Sc. Entrance Exam** Feb 2025
Among 22,000+ candidates (top 0.07%)

25th **National B.Sc. Entrance Exam** Oct 2020
Among 155,000+ candidates (top 0.02%)

Research Experience

Jul 2025 – May 2026
Maximum Divergence Knowledge Distillation
University of Waterloo | Prof. Ghodsi (remote)
- Proposed **MDKD**, a distillation method that uses rejection sampling to focus on outputs the student LLM underestimates.
- Reached state-of-the-art results on **GSM8K**, summarization, and code generation.

Apr 2023 – Feb 2026
Mental Wellness Chatbot
Imperial College London | Prof. Edalat (remote)
- Built a chatbot that recommends exercises based on **Self-Attachment Theory (SAT)**.
- Trained a 12-class **emotion classifier** (92% F1) and fine-tuned GPT-2 with **PPO**.
- Added **RAG** and fine-tuned **Llama-3** on mental-health datasets.

Apr 2025 – Feb 2026
Synthetic Data in Data Selection Methods
Rensselaer Polytechnic Institute | Prof. Amiri (remote)
- Built an **adversarial** data-selection method for pretraining foundation models.
- Analyzed how data-mixing methods such as **DoReMi** weight synthetic data.

May 2024 – Dec 2024
News Importance Classification with LLMs
Sharif University of Technology | Prof. Jafari
- Compared open-source LLMs' **reasoning** on a new human-labeled Persian news dataset.
- Tested prompt sensitivity with **symbol tuning**; designed **query-dependent prompts**.

Jun 2024 – Oct 2024
LLM Citation Benchmark
Sharif University of Technology | Prof. Asgari
- Built a framework and dataset to evaluate the **medical citation quality** of LLMs.
- Scored citations by checking their **facts**, weighted for accuracy and relevance.

Nov 2023 – Sep 2024
Survey on Diffusion Models
Sharif University of Technology | Profs. Rohban & Soleymani
- Studied **compositional** failures in image generation with **diffusion models**.

Industry Experience

Senior ML Engineer | AVIA — Jun 2025 – Present
- Benchmarked **system-one** models (e.g., JEV) on decision tasks and improved **prompt/prefix caching**.
- Built a **LiveKit** VoIP voice-assistant backend and a chat-evaluation dashboard (Django, React).

LLM Engineer | MCINEXT — Oct 2025 – Jun 2026
- Built an **AR-GRPO** training pipeline for agentic LLMs (Gemma3-4B, Qwen3-4B).
- Designed an **adaptive agent** flow with a router model for intent and tool selection.
- Built search and chatbot **evaluation** (LLM judges, win rate), deployed with CI/CD, Helm, and ArgoCD.

AI Engineer | Mohaymen ICT Group — Feb 2025 – Sep 2025
- Built FastAPI services for **NER** (ParsBERT), **translation** (SeamlessM4T), **ASR** search (NeMo), and **OCR** (Tesseract, RolmOCR).
- Sped up the **RAG** system with async and multi-process code, 3-level **Redis** caching, and **vLLM**.
- Ran services with Docker Compose, traced them with Phoenix, and load-tested with Locust.

Machine Learning Engineer | Mohaymen ICT Group — Jul 2024 – Nov 2024
- Fine-tuned **BERT** to find similar questions and built a context-aware **spelling corrector**.
- Built **RAG** with LangChain/LangGraph and improved speech-to-text and text-to-speech with **NeMo**.

Software Engineering Intern | Code Star — Jun 2021 – Sep 2021

- Built **C#** data pipelines and SQL-based ETL.

Teaching

Head TA	*Compiler Design*	Dr. S. Hosseinmardi	Fall 2023, Spring 2024
	Mobile Programming	Dr. S. Nari	Spring 2023
Graduate TA	*Deep Learning*	Dr. M. Soleymani	Fall 2026
	Advanced Computer Vision	Dr. S. Kasaei	Fall 2026
	Natural Language Processing	Dr. M. Samiei	Fall 2026
TA	*Modern Information Retrieval*	Dr. M. Soleymani	Spring 2026
	Mathematics for Artificial Intelligence	Dr. A. Najafi	Spring 2024
	Game Theory	Dr. M. Seddighin	Spring 2024
	Design of Algorithms	Dr. M. Abam	Spring 2024
	Machine Learning	Dr. A. Motahari	Fall 2023, Spring 2024
	Database Design	Dr. M. Akhi	Fall 2023
	Discrete Structures	Dr. M. Abam	Summer 2023
	Data Structures and Algorithms	Dr. M. Ghorbani	Spring 2023
	Numerical Methods	Dr. S. Hosseinghorban	Fall 2022, Spring 2023, Fall 2023
	Probability and Statistics	Dr. M. Motahari	Fall 2022
	Linear Algebra	Dr. S. Hosseinghorban	Spring 2022, Fall 2022, Spring 2023
	Advanced Programming	Dr. V. Salmani	Fall 2021

Community Involvement

Karsoogh Mehregan 23–26 | NODET Alumni Community — Aug 2023 – 2026

Head of Education & Tournament

- Led the science and tournament teams; ran AI and cryptography workshops for teenagers.
- Designed online games that teach machine learning, reinforcement learning, and cryptography.

WebMasters | Sharif University of Technology — Apr 2022

Mentor

- Taught free front-end tutorials (HTML, CSS, JavaScript) to local students.

Skills

Machine Learning	PyTorch, TensorFlow, scikit-learn, Transformers, vLLM, FAISS, Rasa, NumPy, Pandas, Matplotlib, Streamlit
Programming	Python, Java, C#, Swift, FastAPI, Django, REST APIs, PostgreSQL, MongoDB
Web	HTML, CSS, JavaScript, TypeScript, React
DevOps	Docker, Docker Compose, Kubernetes, Redis, ArgoCD, Kibana
Testing & Monitoring	Locust, Phoenix
Research Tools	Zotero, Citavi

Languages

English Professional IELTS 7.5 (R 8.5, L 8.0, S 7.0, W 7.0) | **Persian** Native | **French** Elementary